Kaiyuan (Harry) Deng

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Education

Northwestern University **Sep 2024 – Expected Dec 2025**
Master of Science in Machine Learning and Data Science (ex-Analytics) / ***Cumulative GPA****: 3.94* *Evanston, IL*
Relevant Courses: Machine Learning, Cloud Engineering, Deep Learning, Text Analytics, Data Mining

University of Minnesota - Twin Cities **Sep 2020 – May 2024**
Bachelor of Science in Statistical Science, Minor in Mathematics / ***Cumulative GPA****: 3.93* *Minneapolis, MN*

Technical Skills

Programming: Python (NumPy, pandas, Scikit-learn, Seaborn, PyTorch, TensorFlow, Matplotlib), SQL, R (ggplot2, shiny, dplyr, tidyr), C++, MATLAB, Spark, Hadoop, HTML

Software: Amazon WorkSpaces, Databricks, Tableau, PowerBI, Posit, Excel, Snowflake, GitHub

Professional Experiences

Astellas Pharma (Contracted by Randstad Sourceright) **Jul 2025 – Sep 2025**
Brand Analytics Intern *Northbrook, IL*

- Created an analytical data layer on 1.5M+ pharmacy claims data by joining with Vasomotor symptoms (VMS) and other reference data on AWS Redshift and engineering features (prescriber days-supply preference, patient treatment-history flags etc.) in SQL Workbench, optimizing performance through CTE refactoring and column pruning to cut average runtime by ~35%.
- Conducted Prescriptive Analysis across prescriber and access segments, leveraging Survival & Hazard Tables to quantify conditional conversion rates from 30- to 90-day fills at each prescription event, uncovering market dynamics and adoption drivers.
- Built Decision Tree models to identify physician- and patient-level drivers of 90-day prescribing, revealing prescriber profiles and patient characteristics associated with higher 90-day adoption, and clarifying how these factors shape prescribing behavior.
- Collaborated with marketing team to translate drivers into actionable insights, pinpointing physician segments inclined to prescribe 90-day fills and Managed Care Organizations (MCOs) with higher 90-day adoption to guide access and engagement strategies.

Calamos Investments **Oct 2024 – May 2025**
Data Scientist Consultant - Industrial Practicum *Chicago, IL*

- Built machine learning models (e.g., Logistic Regression, LightGBM, and XGBoost) based on 24 months of customer pre-purchase activity data to predict client deal likelihood in Python, and conducted customer behavior analysis to enable the sales team to develop personalized strategies and increase conversion efficiency.
- Developed KPIs (time-to-conversion metrics etc.) and implemented feature engineering such as activity streak metrics and multi-dimensional interaction to support factor analysis that revealed critical predictors for optimizing customer conversion rates.
- Implemented Elastic Net (LASSO/Ridge hybrid) for feature selection to identify the most predictive variables, complemented by SHAP value analysis to quantify and visualize the impact of meetings, calls, and voicemail interactions on deal closure probability.

Beijing Seeyon Internet Software Corp. **Jul 2024 – Sep 2024**
Data Analyst Intern *Beijing, China*

- Engineered SQL queries in MySQL using CTEs and window functions to conduct RFM segmentation analysis, identifying high-potential accounts and enabling targeted sales outreach strategies that increased retention rates by 25%.
- Built Tableau dashboards on 3M+ community data to visualize key metrics including public safety risks and public service facilities etc., enabling proactive decision-making that reduced incident response time by 40% and improved resource allocation efficiency.

Hongtai Aplus **Jun 2023 – Jul 2023**
Data Analyst Intern *Beijing, China*

- Employed PCA to reduce dimensionality of mutual fund data from 50+ attributes to 12 key components while retaining 84.3% of variance, enabling more efficient clustering analysis of 5k+ funds and improving computational performance.
- Applied GMM and DBSCAN clustering algorithms in Python to categorize mutual funds, improving classification accuracy from 0.65 Silhouette Score (K-means baseline) to 0.82, enabling more precise market positioning and targeted investor outreach strategies.
- Delivered three 40+ page presentations with 15+ PowerBI data visualizations including cumulative asset growth charts, R&D pipeline progression timelines, and patent portfolio distribution maps, translating insights into actionable recommendations for stakeholders.

Project Experiences

Real Estate Lease-Up ML Modeling **Jan 2025 – Mar 2025**

- Optimized processing of 10k+ rows of real estate data across multiple Excel sheets, reducing runtime by 8% while extracting key metrics including post-2008 properties, average lease-up times, and properties with negative rent growth using Python.
- Engineered predictive features for real estate lease-up time modeling, including property competitive positioning metrics, effective age calculations, and lease-up period pricing strategies etc., and implemented multiple regression models (Linear Regression, Random Forest, CatBoost) that achieved ~0.84 R-square, enabling more informed rental price optimization strategies.
- Developed an interactive Streamlit dashboard with GenAI-powered natural language query functionality, enabling stakeholders to instantly visualize market clustering and lease-up analysis, reducing manual analysis processes under 15 mins per market evaluation.